RAJALAKSHMI ENGINEERING COLLEGE

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N2

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|---|---|
| **Started on** | Tuesday, 25 August 2026, 10:22 PM |
| **State** | Finished |
| **Completed on** | Tuesday, 25 August 2026, 10:23 PM |
| **Time taken** | 1 min 6 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int binarySearch(int arr[], int low, int high, int key)
{
    if (low > high)
        return -1;

    int mid = (low + high) / 2;

    if (arr[mid] == key)
        return mid;

    if (arr[mid] > key)
        return binarySearch(arr, low, mid - 1, key);

    return binarySearch(arr, mid + 1, high, key);
}

int findPair(int arr[], int n, int x, int index)
{
    if (index >= n)
        return -1;

    int required = x - arr[index];

    int pos = binarySearch(arr, index + 1, n - 1, required);

    if (pos != -1)
    {
        printf("%d\n", arr[index]);
        printf("%d\n", arr[pos]);
        return 1;
    }

    return findPair(arr, n, x, index + 1);
}

int main()
{
    int n, x;

    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &x);

    if (findPair(arr, n, x, 0) == -1)
        printf("No");
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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